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Operations Research

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Sample solution Endterm SS23

Exam duration: 120 Minuten

Name: ....

Matriculation number: ....

Exercise 1 (19 Points). *Duality*

Problem

Given the following primal LP (P):

$$
\begin{array}{llrcl}
(P) & \underset{\text{s.t.}}{\text{Max}} & -3x_1 + 4x_3 & & \\
 & & +2x_2 - 3x_3 & \leq & 3 \\
 & & -2x_1 + 4x_3 & \leq & 0 \\
 & & 3x_1 - 4x_2 & \leq & -4 \\
 & & x & \in & \mathbb{R}^3_{\geq 0}
\end{array}
$$

a) Derive the dual LP (D) from the primal LP (P).

b) Show that the primal LP (P) is not unbounded.

c) Show that the primal problem (P) and the dual problem (D) have the same feasible region.

d) Show that the optimal objective function value is zero.
Hint: First show, using part c) and weak duality, that the optimal objective function value of the primal problem is non-positive.

Solution

a)

$$
\begin{array}{llrcl}
(P) & \underset{\text{s.t.}}{\text{Min}} & 3y_1 - 4y_3 & & \\
 & & -2y_2 + 3y_3 & \geq & -3 \\
 & & 2y_1 - 4y_3 & \geq & 0 \\
 & & -3y_1 + 4y_2 & \geq & 4 \\
 & & y & \in & \mathbb{R}^3_{\geq 0}
\end{array}
$$

b) To show that the primal problem (P) is not unbounded, it suffices to find a feasible point for (D) (weak duality).
For example, $y = (0, 1, 0)$. Obviously, $y \in \mathbb{R}^3_{\geq 0}$ and furthermore we have

$$
\begin{aligned}
-2 \cdot 1 + 3 \cdot 0 = -2 &\geq -3 \\
2 \cdot 0 - 4 \cdot 0 = 0 &\geq 0 \\
-3 \cdot 0 + 4 \cdot 1 = 4 &\geq 4
\end{aligned}
$$

Additionally, $y \in \mathbb{R}^3_{\geq 0}$, so the point is feasible.

c) Multiplying both sides of the dual constraints by -1 results in the same constraints as in the primal problem (P). Therefore, any feasible solution of the dual is also a feasible solution of the primal and vice versa.

d) Let z^* be the optimal objective value and x^* the primal optimal solution:

$$z^* = -3x_1^* + 0 \cdot x_2^* + 4x_3^*$$

From part (c), we know that the point x^* is also a feasible solution of the dual problem. From weak duality, we obtain the following upper bound for the optimal objective value:

$$z^* \leq 3x_1^* + 0 \cdot x_2^* - 4x_3^* = -z^*$$

This holds for all non-positive numbers, thus we have $z^* \leq 0$. Conversely, for the optimal solution of the dual problem, we can find a lower bound and obtain $z^* \geq -z^*$ or $z^* \geq 0$. From strong duality, it follows that $z^* = 0$.

Exercise 2 (22 Points). *Ice cream production*

Problem

Ice cream maker Theodor wants to create some new ice cream flavors to stand out against the tough competition. He has the basic ingredients in $B = \{1, \dots, m\}$ at his disposal. Additionally, he can enhance each flavor with special ingredients from $S = \{1, \dots, n\}$. He wants to bring a maximum of p new ice cream flavors to the market. Let $K = \{1, \dots, p\}$ denote the set of possible new flavors.

He introduces variables $x_{kb} \in \{0, 1\}$ and $x_{ks} \in \{0, 1\}$ for all $k \in K$, $b \in B$, and $s \in S$, which indicate whether basic ingredient b or special ingredient s is used for flavor k, respectively.

Theodor's objective function is a well-kept business secret. Therefore, only help him in formulating the constraints.

a) Each flavor may contain a maximum of 5 basic ingredients. (3 points)

b) Each flavor must contain at least twice as many basic ingredients as special ingredients. (3 points)

c) Theodor introduces auxiliary variables $P_k \in \{0, 1\}$ for all $k \in K$ to indicate whether flavor k is produced, i.e., if there exists a $b \in B$ such that $x_{kb} = 1$. P_k should be equal to 1 if and only if k is produced. Define the P_k using linear (in)equalities. (4 points)

d) Each produced flavor must contain at least 2 basic ingredients. (3 points)

e) No two flavors may contain more than 2 identical basic ingredients. (6 points)

Solution

a)

$$\sum_{b \in B} x_{kb} \leq 5 \, \forall k \in K$$

b)

$$\sum_{b \in B} x_{kb} \geq 2 \sum_{s \in S} x_{ks} \, \forall k \in K$$

c)

$$P_k \leq \sum_{b \in B} x_{kb} \, \forall k \in K$$
$$M P_k \geq \sum_{b \in B} x_{kb} \, \forall k \in K$$

d)
$$\sum_{b \in B} x_{kb} \geq 2P_k \, \forall k \in K$$

e) For $k, l \in K$ with $k \neq l$ and $b \in B$, introduce the variable $Y_{klb} \in \{0,1\}$ where $Y_{klb} = 1$ if both flavors k and l use ingredient b.

$$Y_{klb} \geq x_{kb} + x_{lb} - 1 \quad \forall k \leq l \in K, \, b \in B$$

Then,
$$\sum_{b \in B} Y_{klb} \leq 2 \quad \forall k \neq l \in K.$$

Exercise 3 (16 Points)**.** *Branch-and-Bound*

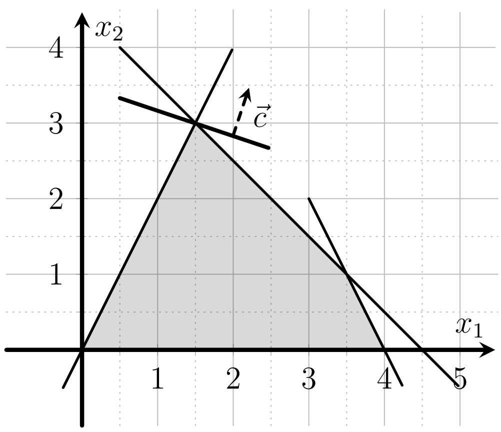
Problem

Given the following integer linear program (P):

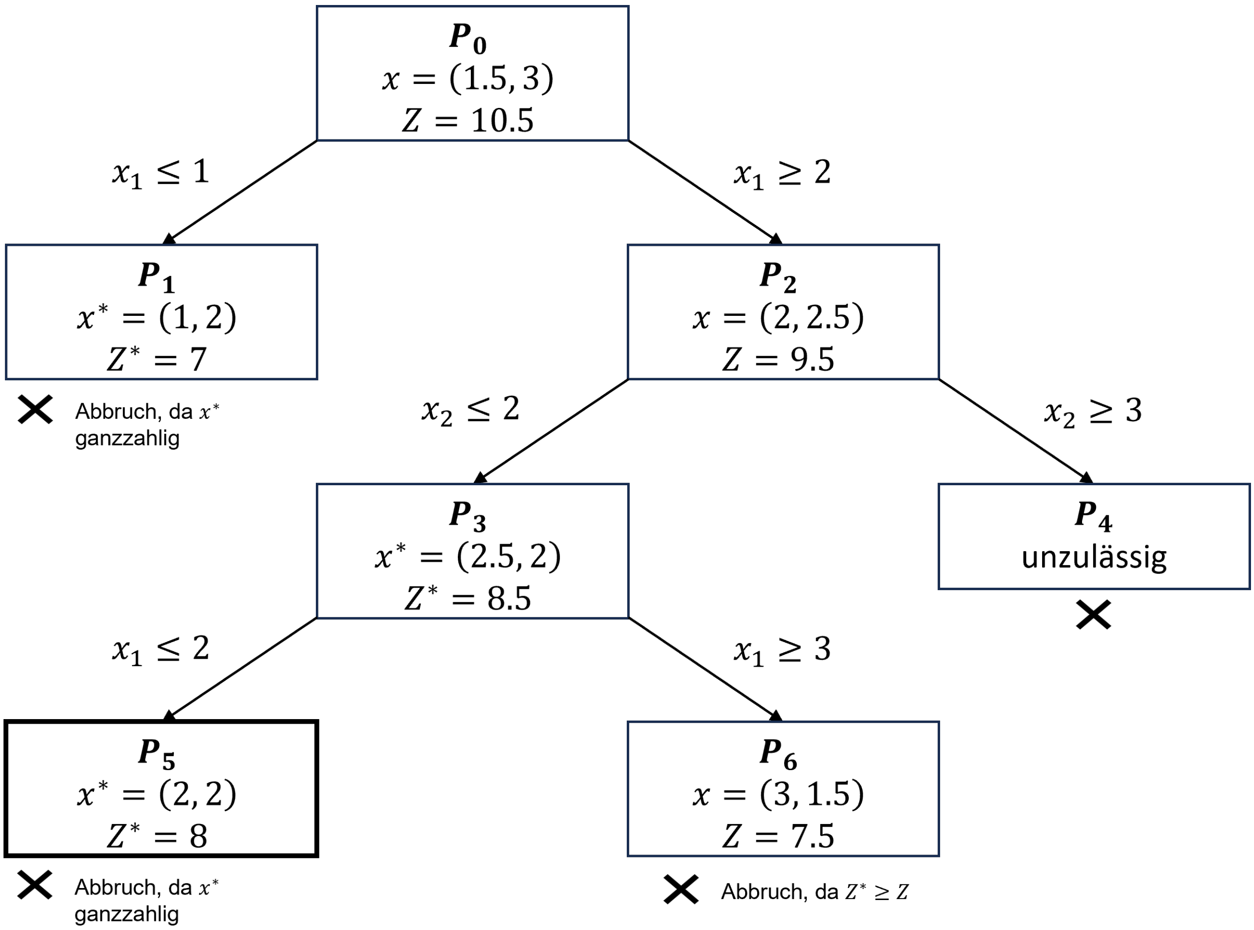
$$\begin{array}{lrcl} (P) \ \underset{\text{s.t.}}{\text{Max}} & x_1 + 3x_2 & & \\ & -2x_1 + x_2 & \leq & 0 \\ & 2x_1 + 2x_2 & \leq & 9 \\ & 2x_1 + x_2 & \leq & 8 \\ & x_1, x_2 & \geq & 0 \\ & x_1, x_2 & \in & \mathbb{Z} \end{array}$$

Solve the integer linear program (P) using the Branch-and-Bound method. Specify the optimal values for x_1 and x_2, as well as the optimal value of the objective function.

Hint: In the following coordinate system, the feasible region of the LP relaxation of (P) as well as the objective function is plotted. You can use this to solve the subproblems graphically.



Solution



The optimal solution to the integer linear programming problem is $x_1 = 2$, $x_2 = 2$ with an objective function value of 8.

Exercise 4 (14 Points). *Collectibles*

Problem

Teodor has decided to sell his old collection of vinyl records as he no longer has a record player. He plans to sell the records at a music lovers' market. Since he doesn't have a car, he must walk to the market and can only carry records with a total weight of up to 5 kg.

Teodor has a rough idea of the price he wants to sell the records for. Here is a list of the records in his collection:

Record	Price in Euros	Weight in kg
The Beatles - "Abbey Road"	2	3
Pink Floyd - "The Dark Side of the Moon"	3	2
Led Zeppelin - "IV"	4	4
Queen - "A Night at the Opera"	1	2
Nirvana - "Nevermind"	2	1

Teodor wants to use dynamic programming (DP) to determine which records he should take to the market to maximize his profit.

a) Teodor knows that for using DP here, he can either break down by *price* or *weight*. Which choice is more efficient? Justify your decision.

b) Execute the DP algorithm with the more efficient choice to determine which records Teodor should take.

c) Which records does Teodor take according to part b)? What is his maximum profit?

Solution

a) One can categorize the use of the DP algorithm either by weight W or by value V. The runtime is $\mathcal{O}(nW_{\max})$ for a breakdown by weight, or $\mathcal{O}(nV_{\max})$ for a breakdown by value, where n is the number of records. Since the potential maximum value $V_{\max} = 12$ is higher than the maximum weight $W_{\max} = 5$, it is better here to categorize by weight.

b) When categorized by weight, using the DP algorithm, we obtain the following table:

Gewicht:	0	1	2	3	4	5
0 vinyl records	0	0	0	0	0	0
1 vinyl records	0	0	0	2	2	2
2 vinyl records	0	0	3	3	3	5
3 vinyl records	0	0	3	3	4	5
4 vinyl records	0	0	3	3	4	5
5 vinyl records	0	2	3	5	5	6

c) From part (b), it is evident that Teodor can achieve a maximum profit of 6 euros. He can either take "Led Zeppelin" and "Nirvana" or "Pink Floyd", "Queen", and "Nirvana".

Exercise 5 (14 Points). *Matroids*

Problem

Given a matroid $M = (E, \mathcal{I})$.

a) Provide the definition of a basis for M.

b) Show or disprove that the bases of M have the same number of elements.

c) Let B_1 and B_2 be two distinct bases of M, with $B_1 \neq B_2$. Show or disprove that $(B_1 \cup B_2) \backslash (B_1 \cap B_2)$ is a basis of M.

Solution

a) A basis is a maximal independent set.

b) We show that the bases of M have the same number of elements.

   Assume that the bases of M do not have the same number of elements. Let B_1, B_2 be two bases of M such that $|B_1| < |B_2|$. Since M is a matroid, there exists an element $e \in B_2 \backslash B_1$ such that $B_1 \cup \{e\} \in \mathcal{I}$. But then $B_1 \cup \{e\}$ is independent and has more elements than B_1, contradicting the maximality of B_1. Therefore, $|B_1| = |B_2|$.

c) We show that $(B_1 \cup B_2) \backslash (B_1 \cap B_2)$ is not a basis of M.

   Consider the matroid $M = (E, \mathcal{I})$ where $E = \{a, b\}$ and $\mathcal{I} = \{\emptyset, \{a\}, \{b\}\}$. Let $B_1 = \{a\}$ and $B_2 = \{b\}$ be bases of M. Then $(B_1 \cup B_2) \backslash (B_1 \cap B_2) = \{a, b\}$. However, $\{a, b\} \notin \mathcal{I}$, hence $\{a, b\}$ is not a basis of M.

Exercise 6 (12 Points). *Modified Nearest-Neighbor-Heuristic*

Problem

Fabienne believes she has found a modification of the Nearest-Neighbor Heuristic that solves the Traveling Salesperson Problem (TSP) optimally under the following assumptions:

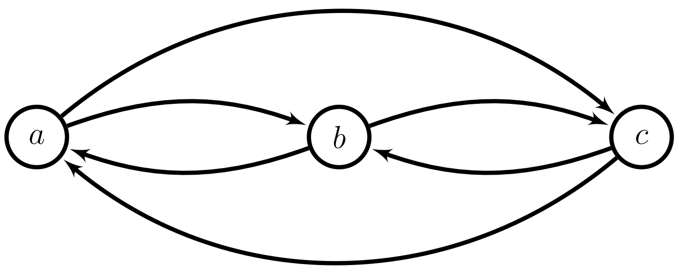
1 The input is a directed graph $G = (V, E)$ with distances $c_{uv} \geq 1, \forall (u, v) \in E$ and a starting node $a \in V$.

2 For all $v \in V \backslash \{a\}$ there is an edge $(v, a) \in E$ with $c_{va} = 1$.

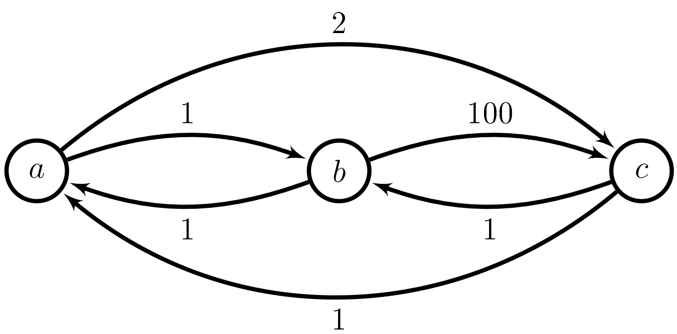
The Modified Nearest-Neighbor Heuristic is as follows:

Start at node $v = a$.
Set of already visited nodes is $V' = \{a\}$.
while $V' \neq V$ **do**
  Find the node $v' \in V \backslash V'$ with the smallest distance from v.
  Add edge (v, v') to the tour.
  Set $v = v'$, $V' = V' \cup \{v'\}$.
end while
Add edge (v, a) to the tour.

Use the following graph to show that Fabienne is wrong. First, assign appropriate weights to the edges. Provide both the optimal solution and the solution calculated with the Modified Nearest-Neighbor Heuristic.



Solution



The optimal solution is $a \to c \to b \to a$ with a total distance of 4. The solution computed with the Modified Nearest-Neighbor Heuristic is $a \to b \to c \to a$ with a total distance of 102.

Exercise 7 (23 Points). *Local Optima*

Problem

Let $\alpha > 0$ be a parameter. Determine all critical points of the function $f_\alpha : \mathbb{R}^2 \to \mathbb{R}$ given by

$$f_\alpha(x, y) = -\frac{1}{4}x^4 - \frac{1}{4}y^4 + \alpha xy \tag{1}$$

as a function of α and show in each case whether it is a local maximum, local minimum, or a saddle point.

Solution

To determine the critical points, we first compute the gradient of the function f_α:

$$\nabla f_\alpha(x, y) = \begin{pmatrix} -x^3 + \alpha y \\ -y^3 + \alpha x \end{pmatrix}. \tag{2}$$

The critical points occur where the gradient equals zero, so we solve $\nabla f(x, y) = 0$. From the first component, we obtain $y = \frac{1}{\alpha}x^3$. Substituting into the second component gives us:

$$-\left(\frac{1}{\alpha}x^3\right)^3 + \alpha x = 0 \Leftrightarrow x\left(-\frac{1}{\alpha^3}x^8 + \alpha\right) = 0. \tag{3}$$

This equation holds when $x = 0$ or $-\frac{1}{\alpha^3}x^8 + \alpha = 0 \Leftrightarrow x^8 = \alpha^4 \Leftrightarrow x^2 = \alpha$. Thus, substituting back into the first equation, the critical points are: $(x_1, y_1) = (0, 0), (x_2, y_2) = (\sqrt{\alpha}, \sqrt{\alpha}), (x_3, y_3) = (-\sqrt{\alpha}, -\sqrt{\alpha})$.

Next, we determine whether these critical points are local maxima, minima, or saddle points. To do this, we compute the Hessian matrix:

$$Hf(x, y) = \begin{pmatrix} -3x^2 & \alpha \\ \alpha & -3y^2 \end{pmatrix}. \tag{4}$$

To assess local optimality, we examine the Hessian matrices at the critical points, starting with:

$$Hf(x_2, y_2) = Hf(x_3, y_3) = \begin{pmatrix} -3\alpha & \alpha \\ \alpha & -3\alpha \end{pmatrix}. \tag{5}$$

Applying the principal minors criterion, we find $\det(-3\alpha) = -3\alpha < 0$ and $\det(Hf(x_2, y_2)) = 9\alpha^2 - \alpha^2 = 8\alpha^2 > 0$. Therefore, the Hessian matrix at (x_2, y_2) and (x_3, y_3) is negative definite, indicating local maxima.

The Hessian matrix at (x_1, y_1) is:

$$Hf(x_1, y_1) = \begin{pmatrix} 0 & \alpha \\ \alpha & 0 \end{pmatrix}. \tag{6}$$

The principal minors criterion does not apply here, so we find the eigenvalues:

$$\det(Hf(0, 0) - \lambda I_2) = \lambda^2 - \alpha^2 = (\lambda - \alpha)(\lambda + \alpha). \tag{7}$$

Thus, the eigenvalues are $\lambda_1 = \alpha > 0$ and $\lambda_2 = -\alpha < 0$, indicating (x_1, y_1) is a saddle point.